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ORIGINAL PAPER

Black Spot Syndrome in ocean surgeonfish: using video-based surveillance to quantify disease severity and test environmental drivers

Cheyenna D. G. de Wit[1] · Pieter T. J. Johnson[2]

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Abstract
Observations of Black Spot Syndrome (BSS), a pigmented dermatopathy in marine fishes, have been increasingly reported in important grazers such as surgeonfish and parrotfish in the Caribbean. This condition has been linked to infection by the trematode parasite, *Scaphanocephalus* spp., although relatively little is known about the environmental drivers of infection and how they vary spatially. This study introduces a non-invasive, video-based method to survey BSS presence and severity in ocean surgeonfish (*Acanthurus tractus*). Application of the approach across 35 coastal sites in Curaçao was used to evaluate the influence of environmental factors on BSS, including longitude, herbivorous fish density, wave energy, depth, nutrient pollution, and inhabited surface area. Of the 5123 fish surveyed between February 2022 and January 2023, 70% exhibited visible signs of BSS, and the average number of lesions per fish increased by ~ fivefold from eastern to western sites along the leeward coastline. Within a site, estimates of BSS severity were broadly consistent between different divers, different reviewers of video footage, and the date of sampling, emphasizing the robustness of the surveillance approach. Analyses of environmental factors indicated that BSS decreased with wave intensity while increasing in association with higher nutrient runoff and fishing pressure. This study provides insight into environmental correlates of BSS severity while highlighting the use of video-based surveillance as a non-invasive survey method. The mechanisms linking environmental factors with BSS as well as its consequences for affected fish remain unknown, emphasizing the need for long-term and experimental studies in this system.

Keywords Host-parasite interaction · Coral reef ecosystem · Disease ecology · Global change · Emerging disease · Non-invasive disease surveillance · Fish disease

Introduction

Despite reported increases in disease syndromes affecting multiple marine taxa (Harvell et al. 2004; Groner et al. 2016; Tracy et al. 2019), parasites remain relatively understudied in marine systems. In most cases, the environmental drivers that contribute to emerging infections are broadly unknown. Some of the most important barriers to studying marine parasites are the inherent logistical and financial challenges to surveying patterns of infection at larger spatial scales, particularly when involving parasites that may be difficult to census owing to their cryptic nature. This emphasizes the need for innovative methodologies to detect and monitor infections, including the use of less-invasive techniques that do not require sacrifice or injury to hosts. Particularly promising approaches include the use of environmental DNA (eDNA) to detect evidence of pathogens from water or sediment and analysis of imagery for disease systems with externally visible pathologies (Amarasiri et al. 2021; Ríos-Castro et al. 2021). For example, the use of repeated, multi-year surveys involving archivable images or video has facilitated a greater understanding of pathogens in marine hosts and their potential drivers (Dalton and Smith 2006;

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✉ Pieter T. J. Johnson
pieter.johnson@colorado.edu

Cheyenna D. G. de Wit
cheyenna.dewit@gmail.com

[1] Freshwater and Marine Biology, University of Amsterdam, Amsterdam, NH, The Netherlands

[2] Ecology and Evolutionary Biology, University of Colorado, Boulder, CO, USA

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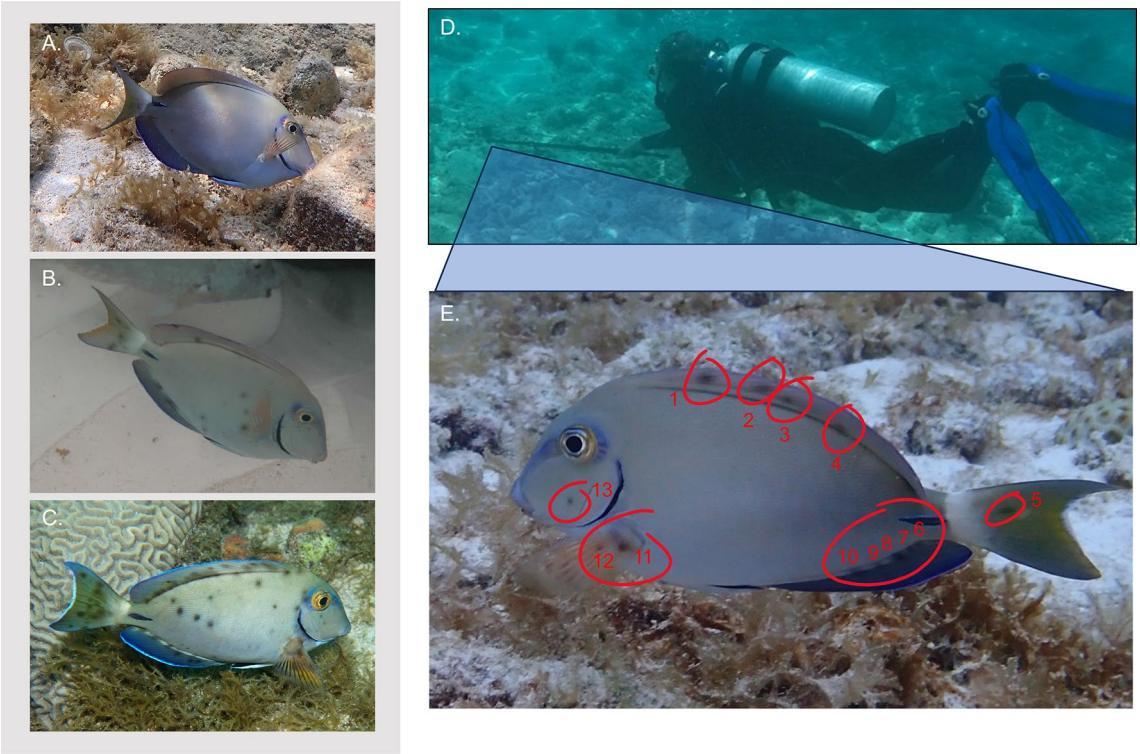

Fig. 1 Black Spot Syndrome in ocean surgeonfish (*Acanthurus tractus*) from Curaçao. The number of spots per fish varies in severity among fish (see panels **A** [3 spots], **B** [18 spots], and **C** [55 spots]). After divers collect video footage during depth-specific transects (**D**), reviewers of the footage quantify the number of spots per fish on one side of each fish (see **E**). Red circles in the last panel are shown to illustrate the counting process. (Images by C. de Wit and P. Johnson)

Lamb et al. 2018; Burns et al. 2020). Lamb et al. (2018) described a multi-species outbreak of ulcerative skin disease in Galapagos fishes, for which bi-annual video monitoring helped link the outbreak to the 2015–2016 El Niño event. Importantly, these techniques can reduce sources of observation bias, the amount of time required underwater, and the taxonomic expertise or training required of observers (see Cappo et al. 2003; Houk and Van Woesik 2006; Mallet and Pelletier 2014; Burns et al. 2020; Castañeda et al. 2020).

In the Caribbean, increasing observations of Black Spot Syndrome (BSS) have been reported in many important fish grazers, such as surgeonfish and parrotfish (Bernal et al. 2015; Elmer et al. 2019). This condition presents as black spots (or pigmented dermatopathies) on the epidermal tissue of the body or fins (Fig. 1) (Dennis et al. 2019). The spots are often well-defined and distinguishable from trauma to the body such as cuts and scrapes (Happel 2019). Using genetic as well as morphological analysis, Kohl et al. (2019) recently linked cases of BSS in the Caribbean to infection by the parasite, *Scaphanocephalus* spp., a group of trematodes with distinctive, wing-like expansions of the anterior region. Importantly, the number of spots visible on affected surgeonfish correlates directly with the number of encysted *Scaphanocephalus* (Kohl et al. 2019), suggesting that non-invasive survey methods—including video-based transects—could provide an effective tool for censusing infection. In locations where BSS has been reported, ocean surgeonfish (*Acanthurus tractus*) often exhibit among the highest frequencies of externally visible infections (Bernal et al. 2015; Dennis et al. 2019; Elmer et al. 2019; Kohl et al. 2019). Although other marine trematodes (e.g., *Cryptocotyle lingua*, *Liliatrema skrjabini*) and other types of parasites (e.g., turbellarians such as *Paravortex* sp.) have been linked to forms of black spot disease in fish (Wakabayashi 1997; Duflot et al. 2023), extensive dissections of reef fishes from Curaçao have identified *Scaphanocephalus* infections as the primary agent, for which loads in individual fish can exceed 500 metacercariae (Malawauw et al. 2024). Like other opisthorchid trematodes, *Scaphanocephalus* spp. have a complex life cycle involving multiple host species across different trophic levels (Kohl et al. 2019). The first intermediate host is likely a marine mollusc, although the specific identity of this host has not yet been established. In the mollusk, parasites reproduce asexually and release free-swimming cercariae that penetrate the skin of fishes as the second intermediate host. In fish, the parasites develop into metacercariae cysts and are eventually eaten by piscivorous birds, for which almost all records to date involve osprey (*Pandion haliaetus*) (Kohl et al. 2019).

Adult parasites reproduce sexually in the gut of the bird, after which the eggs are released with feces of the bird, re-enter the marine environment, and are consumed by mollusks (Kohl et al. 2019). Numerous species of tropical or subtropical marine fish from the Indo-Pacific, Atlantic, Caribbean, and Mediterranean have been reported to support infection by metacercariae of *Scaphanocephalus* spp., including damselfishes, hogfishes, trumpetfishes, goatfishes, wrasse, parrotfishes, razorfishes and pufferfishes (Tubangui 1933; Hutton 1964; Skinner 1978; Inohuye Rivera 1995; Iwata 1997; Bullard and Overstreet 2008; Katahira et al. 2021; Cohen-Sánchez et al. 2023a, b). Emerging evidence suggests there are multiple species of *Scaphanocephalus* that have yet to be fully described either morphologically or genetically (Dennis et al. 2019; González-García et al. 2023), leading us to present only the genus name in this study pending further taxonomic revisions.

Although trematodes in the genus *Scaphanocephalus* have been recorded in numerous fish species, little is known about how infection varies among locations or in response to environmental variables. Elmer et al. (2019) reported that BSS on the island of Bonaire decreased with depth (higher between 2 and 5 m than between 12 and 18 m), and tended to be highest at coastal sites near the urban center of Kralendijk (see also De Graaf and Simal 2015). These patterns likely emerge from a combination of biological and physical characteristics of the environment and their influence on parasite transmission. Previous research on marine trematodes suggests that infection in second intermediate hosts will be a function of the density of infective stages in the environment, the success of those stages in finding a suitable host, and the susceptibility of hosts to becoming and maintaining infection through time (e.g., Combes 2001). The density of infective stages will generally be dictated by the abundance of first intermediate hosts (i.e., snails) and the proportion that are infected (i.e., based on inputs from definitive hosts, such as birds; Fredensborg et al. 2006; Byers et al. 2008). Nutrient runoff and localized eutrophication, which can promote algal growth and snail biomass, lead in some cases to greater production of trematode infective stages and therefore increased infection risk to downstream hosts (see Johnson et al. 2007; Johnson and Carpenter 2008; Vidal-Martínez et al. 2010; Budria 2017). The success of parasites in finding and establishing within suitable hosts is hypothesized to be greater in calm, protected conditions, and can decrease in the presence of predators that reduce infective stage contacts with susceptible hosts (Johnson and Thieltges 2010; Welsh et al. 2017; Koprivnikar et al. 2023). Additional human activities, such as fishing pressure, can have direct and indirect effects on parasite transmission (Lafferty 2008; Wood et al. 2014). Direct harvest of larger-bodied individuals, which tend to be the most heavily infected, can reduce the average infection load in a population; however, top-down losses of predators through overharvesting may lead to increased infections among fish from lower trophic levels owing to greater survival, particularly when those species are not directly fished themselves (Packer et al. 2003; Wood and Lafferty 2015). In summary, spatial variation in parasite prevalence can be driven by environmental variables acting over a hierarchy of scales. Identifying the role of such drivers is essential to better understanding patterns of infection, their potential emergence, and opportunities for management.

In the current study, we set out to optimize a non-invasive, video-based transect method for quantifying the presence and severity of BSS in ocean surgeonfish (*Acanthurus tractus*) between depths and across locations. Our goal was to develop an approach that was easy to implement, generated archivable imagery and was robust to observer variation. We then used this approach to survey reef sites across the leeward coast of Curaçao and assess whether infection patterns varied in response to specific environmental gradients. We focused on ocean surgeonfish because of their high abundance, established susceptibility to infection by *Scaphanocephalus* spp., and the previously demonstrated link between infection and the presence of visibly conspicuous dermatopathies (see Kohl et al. 2019; Malawauw et al. 2024). Surgeonfish are also ecologically important due to their role as grazers on algae that can otherwise adversely affect coral growth (Bellwood et al. 2004; Côté et al. 2013). Building from previous research on marine trematodes, we evaluated the roles of physical and biological variables with the potential to influence exposure and transmission, including wave energy, the amount of coral cover, fish biomass, nitrogen concentrations (as an indicator for sewage runoff), fishing pressure, and the density of nearby houses. These variables, which vary broadly in value across selected sites, also reflect ongoing threats to coral reef ecosystems on the island of Curaçao, such as overfishing, pollution, and losses in coral cover (Jackson et al. 2014; Waitt Institute 2017). How such changes interact to influence emerging infections in marine ecosystems represents an important research frontier.

Methods

Study area

Fieldwork for this study was conducted between February 2022 and January 2023 along the leeward coast of Curaçao (12.1696°N, 68.9900°W) (Fig. 2). Curaçao is an oceanic island in the southern Caribbean Sea located approximately 65 km north of Venezuela. The prevailing currents come from the east/northeast, such that the northern (windward) shore of the island experiences strong wave energy (Van

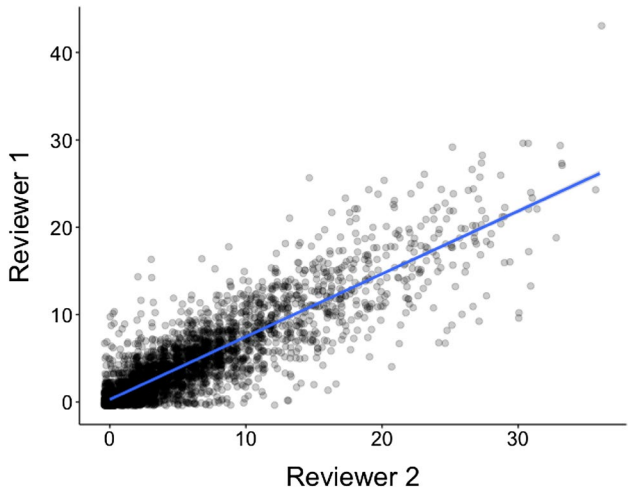

Fig. 2 Concordance between reviewers in quantifying spots per fish among surgeonfish from video transects. For most individual fish ($n = 4495$), two independent reviewers reviewed the footage and estimated the presence and number of spots (lesions). The concordance coefficient between reviewers was estimated as 0.915 (95% CI = 0.905 to 0.924)

den Hoek et al. 1975). The more protected, southwestern coast is bordered by a fringing coral reef that starts with a plateau that slopes down to about 10 m, after which it drops off steeply. Compared to other islands in the Caribbean, Curaçao has relatively high coral cover and fish biomass (Sandin et al. 2008). Nonetheless, factors such as coral cover, fish abundance, and water quality vary considerably among sites along the leeward coastline (Sandin et al. 2022), with generally greater coral cover, higher fish abundance, and less pollution toward the far east side of the island (Waitt Institute 2017). Impacts associated with human development are particularly concentrated around the urban center of Willemstad and the major commercial harbor of St. Annabaai, which collectively represent important sources of runoff for nutrients and other pollutants (e.g., Gast et al. 1999; Klaus et al. 2007).

Study sites and species

Study sites were selected to encompass a wide range of environmental conditions and attributes along the southwestern coast of Curaçao. We identified 35 dive site locations distributed along ~70 km of coastline that captured broad variation in physical and biological characteristics (see Table S1 for a list of sites and locations). At each site, we conducted multiple, replicate transects at both 2- and 5-m depths on the shallow plateau. These depths were selected based on previous research on Bonaire in which the majority of BSS occurred at shallow depths, with a decrease in infection beyond 5 m (Elmer et al. 2019). Surveys focused on the ocean surgeonfish (*Acanthurus tractus*), which is a ray-finned fish in the family Acanthuridae that occurs commonly throughout the Caribbean. It is one of the most abundant herbivorous fish species in the shallow reef habitats around Curaçao (Robertson et al. 2005), and is considered ecologically important in preventing the overgrowth of algae on coral reefs and helping to cycle nutrients (McManus 2000; Burkepile and Hay 2006; Côté et al. 2013; Bloch et al. 2021). Visual counts of BSS were performed on ocean surgeonfish with the use of the video footage collected during transects (see below, and Appendix 1). This species was chosen because of its high abundance and because its pale, uniform coloring facilitates the detection of BSS compared with species exhibiting natural spots or complex color patterns. Previous research has established that *A. tractus* is one of the most externally affected by BSS among studied species (Bernal et al. 2015; Dennis et al. 2019; Elmer et al. 2019; Kohl et al. 2019).

SCUBA transects to assess the severity of Black Spot Syndrome

We used a video-based method to assess BSS presence and severity in *A. tractus* across 35 sites and 2 depths. At each site, SCUBA divers collected video footage using a GoPro® Hero 5 to 8 series attached to a 1-m extension rod, which allowed divers to get closer to individual fish. Three transects were conducted at 2 m and three transects at 5 m. The video resolution was set to 1080 p at 60 frames per second on the linear angle setting (see Appendix 1 for additional details on the sampling and processing protocols as well as sample images). Pairs of divers conducted transects in parallel, with one individual at 2 m depth and the other at 5 m, swimming within eyesight but far enough away to avoid recording the same fish. The divers swam along the depth contour at a speed of approximately 1.5 m per s to locate and record at least 20 haphazardly selected *A. tractus*. Individual fish were approached slowly to obtain footage close enough (~0.5 to 1 m) for subsequent inspection during video analysis. Only adult surgeonfish (estimated at > 15 cm) were recorded. Transects continued for 10 min or until at least 20 ocean surgeonfish had been adequately recorded. Thus, unlike transects focused on quantifying fish abundance (e.g., protocols from the Atlantic and Gulf Rapid Reef Assessment following a static transect line), here divers sought to film as many surgeonfish as possible to provide robust estimates of infection while swimming along a specific depth isobath. After both divers completed a transect at a given depth, they switched depths and conducted another video transect. This

switch was conducted to help minimize observer bias (i.e., each depth was surveyed by multiple divers for each site).

Video processing

Videos were processed using VLC Media Player, a freely available program equipped with viewing tools. During analysis, the video was paused whenever an *A. tractus* was clearly visible and within range for observation. After verifying there were no significant differences in lesion counts on the left versus right side of individual fish, we subsequently counted spots on only one, haphazardly selected side of each fish. For each fish, an observer recorded the estimated number of lesions (Fig. 1) and the time point in the video from which the count was derived. This approach differs from past studies by counting the specific number of lesions rather than classifying BSS as present/absent or into ordinal classes (e.g., De Graaf and Simal 2015; Dennis et al. 2019; Eierman and Tanner 2019; Elmer et al. 2019). When multiple fish were evident in the same frame, we described where each fish occurred spatially (right, left, top, bottom, etc.). If circumstances made it difficult to count the spots accurately (e.g., the fish was too distant, too dark, incompletely in the frame, or at a bad angle), the count was left blank with an explanation as to why no count was made. For a subset of transects, videos were processed by two independent observers; the first observer recorded the time (minutes and seconds) each fish was detected and estimated the number of spots. The second reviewer used the times recorded by the first reviewer but made an independent assessment of the lesion count (i.e., they were not provided with reviewer 1's counts). This approach allowed us to test the congruence of reviewer counts while helping to reduce potential confusion over the identity of the individual fish assessed.

Environmental factors

Data for the site-level environmental factors were derived from existing information for Curaçao collected during an intensive, island-wide survey in 2015 (see Waitt Institute 2017; Sandin et al. 2022). This marine scientific assessment included detailed information on the physical and biological environments at 122 sites along the leeward shore, for which the island was divided into eight zones based on a combination of biological, physical, and anthropogenic characteristics. As part of this survey, Sandin et al. (2022) published an extensive dataset on fish biomass, coral cover, benthic assemblages, fishing pressure, pollution, and site-specific physical features, which was used here to develop potential predictor variables. Previous research on marine parasites has highlighted the influence of both the biotic community and habitat-related differences on infection abundance, including fish abundance, fishing pressure, coral cover, pollution, and water movement (Grutter 1998; Byers et al. 2008; Lacerda et al. 2017; Williams et al. 2022). Building from this foundation, we included the following variables in the analysis as potential predictors of BSS severity (average number of spots detected per fish on one side of the body):

Wave energy

Trematode infective stages emerging from snails (i.e., cercariae) are short-lived with limited swimming ability (Pietrock and Marcogliese 2003), such that their success in contacting suitable second intermediate hosts may be greatest in relatively calm, protected waters (Upatham 1974; Sousa and Grosholz 1991; Galaktionov and Bustnes 1995; Byers et al. 2008). Transmission from bird definitive hosts to susceptible snail hosts may also be greater in calmer, more protected areas, leading to a higher prevalence of infected snails (see Lambert et al. 2012). We used an ordinal measurement of shoreline wave intensity that was ranked through visual assessment and divided into five levels based on wave height: Level 1 (2 to 3.5 m), Level 2 (1.5 to 2.0 m), Level 3 (1 to 1.5 m), Level 4 (0.5 to 1 m), Level 5 (0.3 to 0.5 m), and Level 6 (0 to 0.3 m) (derived from van Duyl (1985)). These values were inverted to make them positively related to wave height (i.e., by multiplying by −1) (Sandin et al. 2022).

Coral cover

The amount of coral cover in shoreline reefs has the potential to inhibit the success of cercariae in contacting fish second intermediate hosts via active predation (i.e., consumption by coral polyps) and through mechanical interference (i.e., structural complexity leads to greater cercariae losses) (Hopper et al. 2008; Paula et al. 2021; Welsh et al. 2024). While corals can function as second intermediate hosts for some trematode parasites (Aeby 2003), this is not the case for *Scaphanocephalus* spp. Coral cover also has the potential to indirectly alter trematode transmission through its effects on the identity and abundance of species in the community, including those directly involved in the parasite's life cycle (e.g., snails and fish) or those that disrupt it (e.g., predators of cercariae or decoy hosts that interrupt transmission). This variable was incorporated as the average percentage of reef-building corals measured using benthic imagery along transects ($n = 122$, depths of 8 to 12 m, see Sandin et al. 2022), followed by a $\log_{10}$-transformation (+1).

Nitrogen concentration

Nutrient runoff can amplify trematode infection through at least two mechanisms. Increased algal growth can promote faster growth and reproduction of aquatic snails, which are the typical first intermediate hosts for trematode parasites (note that the specific first intermediate host of *Scaphanocephalus* remains unknown). In addition, infected snails from eutrophic environments can produce more infective cercariae per day, thereby increasing the exposure risk for local fish populations (e.g., see Johnson et al. 2007; Vidal-Martínez et al. 2010; Budria 2017). Common sources of elevated nitrogen in the nearshore waters of Curaçao include sewage pollution, stormwater inputs, and construction runoff, which have likely contributed to increases in benthic algae and losses of coral (see Gast et al. 1999; Lapointe and Mallin 2011). We used the delta-15-N levels in macroalgae ($n = 122$) (with a $\log_{10}$-transformation (+ 1)) as an indicator of anthropogenic nutrient inputs (see Lapointe and Mallin 2011; Sandin et al. 2022).

Fishing pressure

The intensity of fishing pressure can affect parasite transmission both directly, i.e., by removing large-bodied fish that are often disproportionately infected, and indirectly by reducing predation pressure on unfished species at lower trophic levels (Lafferty 2008; Wood et al. 2014). Because surgeonfish are rarely caught for human consumption or as baitfish in Curaçao, higher fishing pressure is more likely to influence infection indirectly, i.e., by increasing the overall abundance of herbivorous fishes due to reduced predation by piscivorous species or by increasing the abundance of heavily infected fish specifically (which might otherwise by easier prey for predators) (Packer et al. 2003; Wood and Lafferty 2015). The data used for fishing pressure were based on interviews of local anglers ($n = 118$) and subjected to a $\log_{10}$-transformation (+ 1) (see Waitt Institute 2017; Sandin et al. 2022).

Inhabited surface area

Human population density can also influence patterns of parasite infection through the collective effects of disturbance, pollution, and shoreline modification. For instance, human-mediated alterations can inhibit activity by bird definitive hosts, thereby reducing trematode inputs to snails (Smith 2001; Fredensborg et al. 2006; Byers et al. 2008). Alternatively, pollutants can weaken immune defenses or the body condition of fish hosts, leading in some cases to elevated infection levels (Sures and Nachev 2022). As a proxy for overall human activity and density, we used inhabited surface area, which reflects the proportion of the local watershed covered by buildings, commercial areas, and roads (see Sandin et al. 2022).

Fish biomass

Host density or biomass is often a key variable dictating parasite transmission. For trematode cercariae, a higher density of hosts is expected to increase the per capita success of parasites in finding a host while diluting the number of encysted metacercariae detected per fish (Stumbo et al. 2012; Buck and Lutterschmidt 2017). Over longer time periods, however, higher fish density could promote overall parasite transmission and lead to increases in infection pressure (Johnson et al. 2013; Buck et al. 2017). For this measure, we used site-specific estimates of herbivorous fish biomass as derived from transects conducted along the forereef ($n = 122$, 8 to 12 m depth, 300 m^2 of survey area; see Sandin et al. 2022). This choice was made based on the observation that *Scaphanocephalus* appears to infect a wide range of herbivorous fish (e.g., parrotfish, surgeonfish, damselfish).

Statistical analysis

To assess consistency between independent reviewers of video footage, we calculated the concordance correlation coefficient between their spot counts for fish scored by both reviewers. Corcordance analysis measures agreement between alternative methods (or reviewers) to evaluate consistency (e.g., Lin 1989). Because the count of spots per fish involved discrete data that were overdispersed (i.e., the variance was much larger than the mean), we used the R package iccCounts to estimate the intraclass correlation coefficient between reviewers (Carrasco 2010, 2022). This package applies generalized linear mixed models to model the within-subjects variance using alternative discrete distributions (e.g., Poisson and negative binomial) and varying forms of zero-inflation. After identifying the model distribution with the lowest AIC, we tested its validity by comparing randomized quantile residuals from the fitted model with those obtained by simulation (Carrasco 2022).

For subsequent analyses aimed at evaluating patterns of BSS severity among locations, we focused on the scores from reviewer 1 (i.e., only a subset of videos was reviewed by two independent reviewers). Once again, we used generalized linear mixed models (GLMMs) for which we could vary the distribution (Poisson and negative binomial) and incorporate zero-inflation. Models were constructed using the glmmTMB package (Brooks et al. 2017) in the R statistical environment (R Core Team 2023 version 4.3.1). In all models, site identity and the individual transect were included as random intercept terms. As additional assessments of whether the transect method was robust, we

tested how diver identity and date of sampling influenced estimates of BSS severity by including them as categorical fixed effects. These terms were assessed using a likelihood-ratio test and subsequently removed after being found to have no significant influence.

To evaluate how potential drivers of trematode infection affected BSS, we incorporated transect depth (2 m versus 5 m), wave energy, hard coral cover, herbivorous fish biomass, nitrogen concentration, fishing pressure, and inhabited surface area as fixed effects. Prior to inclusion, numeric predictor variables were centered and scaled using the scale function and tested for collinearity to ensure that all pairwise correlation coefficients were less than 0.6. After initial explorations indicated that a zero-inflated model with a negative binomial distribution ('nbinom1') had the lowest AIC score, we used a backward elimination approach to identify terms influential in predicting spots per fish. Beginning from the full model, likelihood ratio tests were applied to sequentially compare against a reduced model with the least significant term removed. Model reduction continued until any further removals resulted in a P-value of <0.1. As diagnostics on the final model, we used the performance package (Lüdecke et al. 2021) to calculate the marginal and conditional R^2 values (i.e., the coefficient of determination considering only the fixed effects [marginal R^2] or both fixed and random effects together [conditional R^2]), variance inflation factors (VIFs, an indicator of collinearity between predictors), overdispersion, and any influential outliers. We also examined whether residuals showed evidence of spatial autocorrelation by calculating their correlation with longitude.

Results

Overview and validity of video transect approach

In total, we examined 5968 ocean surgeonfish on 35 sites along the southern coast of Curaçao. This included 214 transects (105 at 2 m and 109 at 5 m), with an average of 5.2 transects per site. On average, we recorded 26.3 fish per transect over a 5.8-min average time period. Spot counts could not be reliably quantified on 845 fish (14.2%) that were out of frame, too far away, too dark, or at a poor angle. Of the 5123 fish with reliable counts, 3580 (70%) showed visible signs of BSS (one spot or more; range: 1 to 55 spots on one side of the body) (Fig. 1). Spatially among sampled sites, there was a significant effect of longitude on BSS severity, such that the average number of spots per fish increased substantially from east to west along the leeward shore (negative binomial GLMM: scale (longitude) $= -0.435 \pm 0.068$; $z = -6.32$, $P < 0.00001$) (Fig. 2). The lowest values were from East Point (Secrets; $\bar{x} = 0.2$ spots per fish), while the greatest infection was at Rif St. Marie ($\bar{x} = 6.6$ spots per fish), with multiple sites toward West Point exceeding 5 spots per fish (see Fig. 2). There was no effect of depth (2 vs. 5 m) on spot count (scale (depth) $= -0.030 \pm 0.021$; $z = -1.40$, $P = 0.16$; Fig. 2), and the addition of diver identity (as a fixed effect) was not significant (likelihood ratio test comparing model with and without diver identity; $\chi^2 = 25.7$, df $= 19$, $P = 0.14$).

Based on the concordance analysis, the estimated intraclass correlation coefficient between spot counts for different reviewers was 0.915 (95% CI = 0.905 to 0.924). This indicates that, among the subset of fish with multiple assessments, ratings of the two independent reviewers were broadly consistent (see Fig. 3). The best-fitting model used a Poisson distribution and yielded a mean lesion count of 4.54 per fish with a variance among subjects (i.e., individual fish) of 1.65 and a variance between methods (i.e., reviewers) of 0.032. Exponentiation of the fixed effect of reviewer (0.253)

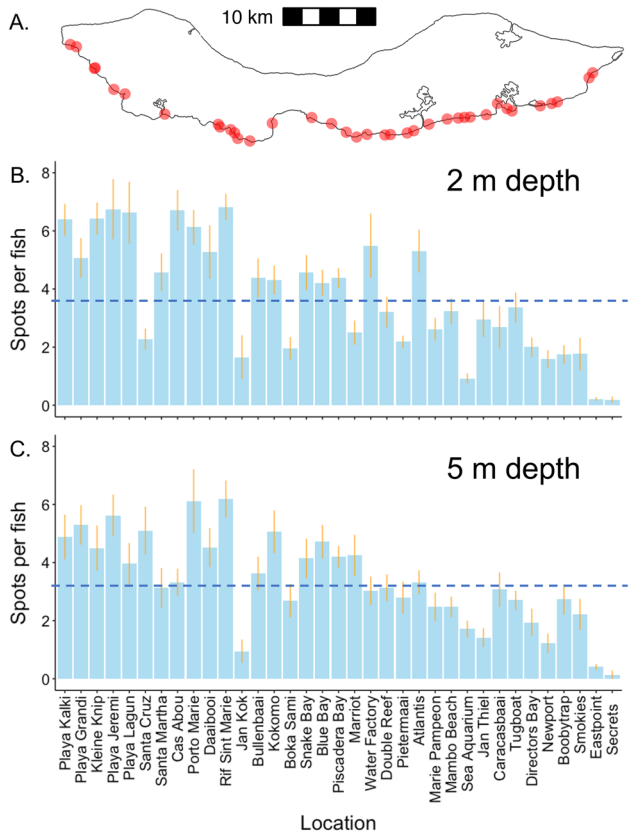

Fig. 3 Variation in Black Spot Syndrome in ocean surgeonfish across sites and depths in Curaçao. Among the 35 sites sampled along the southern coastline (see red circles in map panel **A**), BSS severity (average spots per fish ± 1 SE) varied from < 1 near the eastern edge of the island to upwards of 7 farther to the west. No overall difference was detected between transects conducted at 2 m depth (**B**) versus those at 5 m (**C**) ($n = 2954$ and 2145 fish examined, respectively). Dashed horizontal lines represent the average counts per fish for each depth

from the GLMM yielded a value of 1.28; thus, for every 1 unit increase in lesions reported by reviewer 1, the spot count for reviewer 2 increased by 1.28. This indicates that reviewer 2 consistently counted more lesions than reviewer 1. Fitting the data with a negative binomial model yielded nearly identical results, albeit with a slightly higher AIC value (+2). Incorporation of zero-inflation led to failures in model convergence. Because of the high concordance between reviewers, we subsequently focused on ratings from reviewer 1 only, which provided a larger total sample size (628 fish lacked ratings from reviewer 2).

Effect of environmental factors on the spatial distribution of infection around Curaçao

Based on AIC comparisons of alternative response variable distributions, a model with a zero-inflated negative binomial distribution (nbinom1, or the 'alternative formulation' in glmmTMB had a delta AIC of 5 to 10,628 units lower relative to alternative responses involving Poisson, overdispersed Poisson, or nbinom2). We did not include longitude in this model because it was collinear with several of our hypothesized predictors (Pearson $r > 0.7$), and because the goal was to better understand what factors mechanistically contributed to the observed spatial pattern among sites. Following a backward selection approach, the final reduced model included significant effects for wave energy, fishing pressure, and nitrogen concentration (Fig. 4). Wave intensity negatively predicted spot count per fish (negative binomial GLMM, scale (wave intensity) = -0.367 ± 0.079; $z = -4.65$, $P < 0.00001$), while nitrogen (scale (15N) = $0.312 \pm 0.0.088$; $z = 3.57$, $P = 0.00036$) and fishing pressure (scale (logfishing) = 0.225 ± 0.09; $z = 2.48$, $P = 0.013$) both had positive effects. For instance, wave intensity at East Point sites where infection was lowest averaged -3.3 compared with average values of -5.7 at the west end where infection was higher (Fig. 4). There was no effect of transect depth. The conditional R^2 (fixed and random effect influence) of the final model was 0.25 while the marginal R^2 (fixed effects only) was 0.13. Model diagnostics did not detect evidence of overdispersion ($P = 0.33$) or collinearity (all VIFs < 1.7). Residuals from the final model were weakly correlated to longitude ($r = -0.03$), suggesting incorporated predictor variables broadly accounted for the observed spatial pattern.

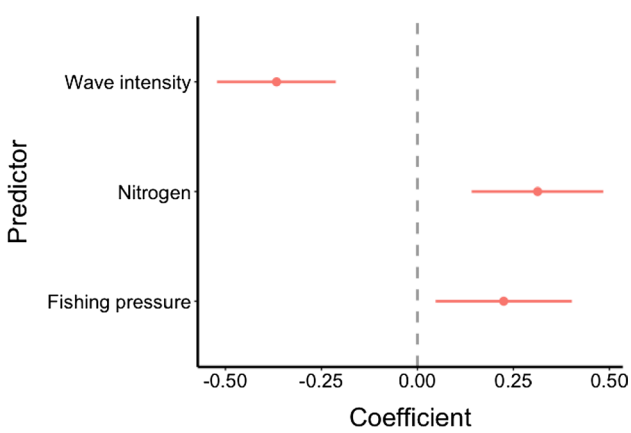

Fig. 4 Influence of environmental variables on BSS severity among sites. Depicted is a coefficient plot illustrating the effect ± 1 SE of terms retained in the best-fitting model predicting BSS counts per fish across all transects and sites. Predictor terms were centered and scaled prior to inclusion to facilitate the comparison of effect sizes. The vertical dashed line at zero indicates a lack of any effect

Discussion

Despite evidence of Black Spot Syndrome (BSS) from archival imagery of *A. tractus* as far back as 1985 (Elmer et al. 2019), we know relatively little about how BSS varies spatially among reef locations and its potential links to environmental factors. A key prerequisite to this inquiry is the development of consistent and repeatable survey methodologies. While many parasitic infections of fish require dissection or molecular-based sampling to detect and quantify, infection by the trematode *Scaphanocephalus* spp.—the primary etiological agent associated with BSS in this region—can be counted via the conspicuous black lesions formed on the otherwise pale bodies of ocean surgeonfish (Kohl et al. 2019). Building from this foundation, here we optimized the use of non-invasive and low-cost video transects conducted during SCUBA dives to quantify variation in BSS at specific depths along the leeward coast of Curaçao. Individual transects were relatively short in duration (~ 10 min) and could be performed by divers swimming in parallel yet sufficiently spaced to avoid filming the same individual fish (although note that this may depend on water clarity and safety concerns). Perhaps most importantly, video footage provides a permanent record that can be analysed by multiple, independent observers and archived for additional comparisons in the future, thereby helping to minimize methodological differences and measurement error among studies or observers. When analysing the footage, for instance, reviewers can pause or replay the video to carefully examine individual fish, which will often be more consistent than attempting to classify spots while diving (including opportunities for post-processing of videos). Concordance analysis of our video transects showed high agreement between two independent reviewers, with an estimated intraclass correlation coefficient of 0.92. By including replicate transects at each depth and site, we were also able to show that the identity of the diver

who conducted the transect did not affect estimates of BSS severity. These results highlight opportunities to extend and apply such video-based transects to evaluate patterns of BSS among islands and geographic regions, with application potential for management and conservation organizations. Community-based science by volunteer networks, including the recreational diving industry, offer additional avenues for obtaining video or photographic material and crowd-sourcing spot counts (e.g., Barve 2014; Daume 2016; Elmer et al. 2019).

Spatial distribution of BSS around Curaçao and insights into potential drivers

Across sites and transects, we detected a high overall prevalence of BSS (~70%) with considerable spatial variation in the average number of spots per surgeonfish around Curaçao. Prevalence and intensity of BSS were lowest in the southeast and highest toward the west end of the island. Longitude was therefore a strongly influential predictor in initial statistical models. On the east end of Curaçao, which is privately owned and associated with less urban and industrial development, BSS was present but the average number of spots per fish was often <2. Infection values increased from east to west along the island, with several western sites exhibiting average spot counts of 6 or higher (maximum count per fish: 55). Similar patterns in BSS distribution were reported on the neighbouring island of Bonaire (see Elmer et al. 2019). Using an ordinal classification system (range: 0 to 4), they found that BSS was lowest along the southern shores and increased around the city of Kralendijk and the northwestern portion of the island. The southeastern shores of both Curaçao and Bonaire are characterized by low anthropogenic development and strong offshore tradewinds, which could drive variation in BSS severity either directly or indirectly. The Bonaire study also reported a significant decrease in BSS with transect depth, which was not detected here. However, this is likely because we included only two depths—2 and 5 m—whereas Elmer et al. (2019) surveyed at 2, 5, 12 and 18 m. They noted a substantial decrease in average infection among the deepest depths (12 and 18 m), which is consistent with the expectation that trematode exposure of fish is associated with nearshore sources. We selected only the two shallowest depths based on this finding as a way to optimize our survey time.

To better understand why BSS severity varied along Curaçao's leeward shoreline, we tested the influence of variables hypothesized to affect trematode infection or transmission. Surprisingly few studies have evaluated potential drivers of heterogeneity in trematode loads among fish on tropical reefs (Cribb et al. 2001; Williams et al. 2022). Based on a model selection approach, we found that the number of lesions per surgeonfish was negatively associated with wave intensity and positively associated with nitrogen concentration and fishing pressure. No statistically significant effects were detected for hard coral cover, nearshore housing cover, herbivorous fish biomass or transect depth. Wave energy, which is generally greatest toward the eastern portion of the island subjected to tradewinds and the open ocean (van Duyl 1985), has the potential to adversely affect transmission between infective cercariae and suitable fish hosts. Although cercariae have tails and are free-swimming, they are generally short-lived (<24 h) and unable to overcome significant water movement, such that calm, protected waters are advantageous for successful contact between infective stages and aquatic hosts (Combes 2001; Lambert et al. 2012). For instance, higher water velocity can damage trematode cercariae and inhibit infection of subsequent hosts (see Upatham 1974; Sousa and Grosholz 1991), as also found for some monogenean, fungal, and myxosporean parasites in aquatic systems (Barker and Cone 2000; Bodensteiner et al. 2000; Hallett and Bartholomew 2008). This same pattern is likely evident in Bonaire. A comparison of wave intensity scores for sites on the island by van Duyl (1985) suggests an inverse relationship with the BSS scores presented by Elmer et al. (2019), similar to that reported here. Wave energy is also very high on the north shore of Curaçao, which unfortunately precluded us from collecting data at such sites in the current study. Future surveys of the north shore conducted during calmer times of the year would provide a valuable comparison.

Nutrient pollution had a positive influence on the observed severity of BSS. As an indicator of sewage and domestic runoff, isotopically heavy nitrogen (delta-15-N) within algal tissue tends to be highest in the waters near the main city of Willemstad and lowest on less-populated eastern and western ends of the island (Klaus et al. 2005; Lapointe and Mallin 2011; Sandin et al. 2022). Runoff of nutrients and other pollutants can enhance parasite infection through multiple mechanisms. Contaminant exposure can reduce immunological defenses and increase host susceptibility to trematode infection (Sures and Nachev 2022). Eutrophication associated with coastal runoff of N and P from wastewater elevates algal growth and can increase the biomass of the herbivorous snails that serve as trematode intermediate hosts. Although the specific intermediate host for *Scaphanocephalus* spp. is not yet known, it is likely to be a snail found in coastal or shallow nearshore environments. In an experimental study, Johnson et al. (2007) showed that eutrophication caused both an increase in snail density and greater per-snail production of infective cercariae, which collectively elevated infections among second intermediate amphibian hosts by ~fivefold relative to non-eutrophic (ambient) conditions. Other studies

have similarly linked nutrient runoff with increased helminth infections in both freshwater and marine ecosystems (Johnson and Carpenter 2008; Johnson et al. 2010; Vidal-Martínez et al. 2010; Budria 2017). However, high levels of eutrophication and other contaminants can also inhibit trematode infection or the availability of hosts on which their life cycles depend, emphasizing the importance of additional research to understand the mechanisms underlying observed associations (e.g., see Blanar et al. 2009).

Finally, higher fishing pressure also correlated positively with the average spot count per surgeonfish. Previous, comparative work in marine environments has found that fishing can be associated positively or negatively with parasite abundance, depending on transmission strategy (e.g., direct vs. multi-host life cycles) and the specific identity of the parasite. Wood et al. (2014) reported that while the abundance of many multi-host parasites decreased with higher fishing pressure, responses of trematodes were highly variable among species. For metacercariae of one trematode species that covaried positively with fishing pressure (i.e., *Stephanostomum* sp.), the authors hypothesized that harvest of predatory fish may release intermediate snail hosts from predation, thereby increasing the density of snails able to transmit trematode infective stages. How fishing affects trematode infections is likely to depend strongly on the hosts in the life cycle: for trematodes that use predatory fish as definitive hosts, high fishing pressure—which often targets higher trophic levels—is likely to lower host availability and reduce infection abundance. But for trematodes that use fish from lower trophic levels as second intermediate hosts and mature in fish-eating birds, as in the case for *Scaphanocephalus* spp., human-associated harvesting may increase host availability (i.e., by decreasing predation pressure on fish lower in the food web). Further comparisons both among islands and across locations with different fishing practices will help to understand the mechanistic links underlying this relationship. In a study from the Mediterranean, for instance, Cohen-Sánchez et al. (2023a) reported higher average loads of *Scaphanocephalus* spp. metacercariae from razorfish (*Xyrichtys novacula*) outside of a marine protected area than within its borders, which the authors suggested could stem from differences in either parasite exposure or stress-mediated changes in host susceptibility.

This study characterized spatial variation in BSS severity among ocean surgeonfish from Curaçao and identified environmental factors that may help explain such patterns. Given that these relationships are correlational, it is important to recognize that other variables might additionally or alternatively be involved. For instance, if the distribution of fish-eating osprey—which are the only known definitive hosts of *Scaphanocephalus* in the Caribbean—mirrors that of human fishing pressure, observed spatial variation may emerge from differences in bird activity and its role in spreading parasite propagules, rather than any influence of human harvest (e.g., Fredensborg et al. 2006; Byers et al 2008). The additional movement of birds, fish, and water among sites all have the potential to alter the observed spatial pattern in ways that could alter or obscure the underlying mechanisms. These challenges are exacerbated by ongoing uncertainty about the identity of the first intermediate host for *Scaphanocephalus*, which so far has precluded additional comparisons between fish and infection within this source host. The high overall prevalence of BSS among surgeonfish reported in this study raises additional questions about the consequences of infection for individual fish hosts and how these might 'scale up' to affect the coral reef ecosystem. For example, Cohen-Sánchez et al. (2023a, b) reported that *Scaphanocephalus* loads in razorfish were positively associated with markers of oxidative and immunological stress while negatively correlated with fish body condition. Future research to examine these questions—along with investigations into seasonal changes in the abundance of migratory osprey—would greatly help to advance this field. We highlight the utility of non-invasive, video-based transect methods for studying marine disease phenomena and hope to encourage other research groups, student courses, and community scientists to apply such tools in developing a larger-scale understanding of BSS in affected regions, including the Caribbean, the western Mediterranean, the Arabian Gulf, and Japan (Al-Salem et al. 2020; Shimose et al. 2020; Katahira et al. 2021; Cohen-Sánchez et al. 2023a, b).

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Author contributions PTJJ and CDGW designed the study; CDGW and PTJJ collected the field; PTJJ and CDGW performed statistical analyses and visualizations; CDGW and PTJJ jointly wrote and edited the manuscript.

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org/10.6084/m9.figshare.25471978.v1, https://doi.org/10.6084/m9.figshare.25472086.v1.

Declarations

Conflict of interest The authors declare no relevant financial or non-financial competing interests.

Ethical approval All applicable international, national, and institutional guidelines for the sampling, care, and use of organisms were followed, including necessary approvals.

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